

Require: Input w , LLM π , depth limit D_{max} , sample limit N .
Initialize $E^0 \leftarrow \text{Extract entities on } x$, $P \leftarrow []$,
 $M \leftarrow 0$.
while $D \leq D_{max}$ **do**
 $R_{cand}^D, P_{cand} \leftarrow \text{Search}(x, E^{D-1}, P)$
 $R^D, P \leftarrow \text{Prune}(\pi, x, R_{cand}^D, P_{cand})$
 $E_{cand}^D, P_{cand} \leftarrow \text{Search}(x, E^{D-1}, R^D, P)$
 $E^D, P \leftarrow \text{Prune}(\pi, x, E_{cand}^D, P_{cand})$
if Reasoning(π, x, P) **then**
Generate(π, x, P)
break
end if
Increment D by 1.
end while
if $D > D_{max}$ **then**
Generate(π, x)
end if

Require: Input w , LLM π , depth limit D_{max} , sample limit N .
Initialize $E^0 \leftarrow \text{Extract entities on } x$, $P \leftarrow []$,
 $M \leftarrow 0$.
while $D \leq D_{max}$ **do**
 $R_{cand}^D, P_{cand} \leftarrow \text{Search}(x, E^{D-1}, P)$
 $R^D, P \leftarrow \text{Prune}(\pi, x, R_{cand}^D, P_{cand})$
 $E_{cand}^D, P_{cand} \leftarrow \text{Search}(x, E^{D-1}, R^D, P)$
if Reasoning(π, x, P, E_{cand}^D) **then**
Generate(π, x, P, E_{cand}^D)
break
end if
 $E^D, P \leftarrow \text{Random_Prune}(E_{cand}^D, P_{cand})$
Increment D by 1.
end while
if $D > D_{max}$ **then**
Generate(π, x)
end if

B ADDITIONAL ABLATION STUDY AND EXPERIMENT ANALYSIS

In this section, we conduct more experiments for ablation study in addition to Section 3.2.3, and analyze experimental results of ToG in detail.